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Marketing Strategy Innovation via AI Adoption: A Study on Bangladeshi SMEs in the Context of Industry 5.0

Tahsina Khan*, Md Mehedi Hasan Emon[†], Sadia Rahman[†]

*Centre for Higher Studies and Research, Bangladesh University of Professionals (BUP), Bangladesh

[†]Faculty of Business Administration, American International University-Bangladesh, Bangladesh

Email: tahsina171@gmail.com, emonmd.mhasan@gmail.com, sadiarahm@gmail.com

Abstract— This study investigates the impact of Artificial Intelligence (AI) adoption on marketing strategy innovation among small and medium enterprises (SMEs) in Bangladesh, within the context of Industry 5.0. The research aims to identify the factors influencing AI-driven marketing innovation and how these factors contribute to strategic advancements. A quantitative research design was employed, collecting data from 165 SMEs through a structured questionnaire. Structural Equation Modelling (SEM) using SmartPLS was applied to examine the relationships between the independent variables— Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Industry 5.0 Readiness (I5), Organizational Support (OS), Competitive Pressure (CP), and Cost of AI Implementation (CAI)— and the dependent variable, Marketing Strategy Innovation (MSI). The findings revealed that PEOU, I5, OS, and CP positively influence marketing strategy innovation, while PU was found to have an insignificant impact. Additionally, CAI negatively impacted innovation, though SMEs with strong organizational support and external pressures still innovated despite cost concerns. These results underscore the importance of organizational readiness and user-friendly AI systems in driving innovation. The study's limitations include a focus on Bangladeshi SMEs, which may limit generalizability. Further research is needed across different regions and industries. This research offers insights for SME managers and policymakers to focus on improving AI accessibility and support structures to foster marketing innovation. By highlighting the role of AI in marketing innovation, the study emphasizes the need for technological upskilling and capacity building in developing economies. This study contributes to the literature by exploring AI adoption in the context of Industry 5.0 within Bangladeshi SMEs, addressing a gap in research on AI-driven innovation in developing economies. The limitations of the study include generalizability to other industries and regions, as well as a limited sample size.

Keywords— Artificial Intelligence, Marketing Strategy Innovation, SMEs, Industry 5.0, Bangladesh, Structural Equation Modelling